

E.G.S. PILLAY ENGINEERING COLLEGE

(Autonomous)

NAGAPATTINAM–611002.

(Affiliated to Anna University, Chennai| Accredited by NAAC with 'A++'Grade Accredited by NBA| Approved by AICTE, NewDelhi)



REGULATIONS - R2023

B.E – BIO MEDICAL ENGINEERING

SECOND SEMESTER

COURSE CODE	COURSENAME	CATEGO RY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
Theory Courses									
2301MA202	Calculus, Ordinary Differential Equations And Complex Variables	BSC	3	1	0	4	40	60	100
2302EC201	Electron Devices	PCC	3	0	0	3	40	60	100
2301GEX05	Applied Digital Logic and Design	ESC	3	0	0	3	40	60	100
2301GEX02	Engineering Graphics	ESC	2	1	0	3	40	60	100
2301GEX03	Problem Solving using C	ESC	2	0	4	4	50	50	100
2304FLX02	Language elective-Japanese	EEC	2	0	0	2	100	0	100
2301TA201	Tamils and Technology/ தமிழரும் தொழில்நுட்பமும்	HSMC	1	0	0	1	100	0	100
Laboratory Courses									
2301GEX51	Computer Practices Laboratory	ESC	0	0	2	1	100	0	100
2302EC251	Electron Devices Laboratory	PCC	0	0	2	1	100	0	100
2301GEX54	Applied Digital Logic and Design Laboratory	ESC	0	0	2	1	100	0	100
2301LS201	Life Skills–II	-	0	0	0	0	100	0	100
TOTAL			16	2	10	23	770	230	1100

2301MA202	CALCULUS, ORDINARY DIFFERENTIAL EQUATIONS AND COMPLEX VARIABLE (Common to ECE and BME)							L	T	P	C	
								3	1	0	4	
PREREQUISITE:												
	1.Differentiation											
	2.Integration											
COURSE OBJECTIVES:												
1. To develop the use of Laplace transform is needed by engineers for practical applications..												
2. To find the solution of ordinary differential equations as most of the engineering problems are characterized in this form.												
3. To familiarize the student with analytic functions of complex variables. This is needed in many branches of engineering.												
4. To acquaint the student with mathematical tools needed in evaluating complex integrals and their applications.												
COURSE OUTCOMES:												
On the successful completion of the course, students will be able to												
CO1:	Apply Laplace transforms to solve physical problems arising in Engineering.											
CO2:	Solve engineering problems by using the concepts of gradient, divergence, and curl.											
CO3:	Solve the higher order differential equations using various techniques											
CO4:	Make use of differentiation formulas to construct analytic functions related to complex variable											
CO5:	Apply the concepts of integration for complex functions in certain regions to determine real integrals.											
Cos Vs Pos MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	
CO1	3	2	1	1	1						1	
CO2	3	2	1								1	
CO3	3	2	1		1						1	
CO4	3	2	1								1	
CO5	3	2	1								1	
Cos Vs PSOs MAPPING												
				COs	PSO1	PSO2	PSO3					
				CO1		1						
				CO2		1						
				CO3		1						
				CO4		1						
				CO5		1						
COURSE CONTENTS:												
Module I	LAPLACE TRANSFORM									9+3Hours		
Laplace transform-Inverse Laplace transform-properties of Laplace transforms-Laplace transforms of unit step function, impulse function and periodic function - Convolution theorem - Solution of ordinary differential equations with constant coefficients and system of linear differential equations with constant coefficients using Laplace transform-Applications to electrical circuits.												
Module II	VECTOR CALCULAS									9 +3Hours		
Scalar and Vector fields - Vector Differentiation - Level surfaces - Directional derivative - Gradient of a scalar field - Divergence and Curl of a vector field. Line, surface and volume integrals; Green's theorem in a plane - Gauss Divergence theorem and Stoke's theorem.												

Module III	ORDINARY DIFFERENTIAL EQUATIONS	9 +3Hours
Higher order linear differential equations with constant coefficients–Cauchy’s and Legendre’s linear differential equations-Method of variation of parameters.		
Module IV	COMPLEX VARIABLE – DIFFERENTIATION	9 +3Hours
Analytic functions – Cauchy - Riemann equations (excluding proof) – Properties of analytic function–Harmonic conjugate – Construction of analytic function by Milne Thomson method –Bilinear transformation.		
Module V	COMPLEX VARIABLE–INTEGRATION	9+3 Hours
Cauchy’s integral theorem-Cauchy’s integral formula for derivatives – Cauchy residue theorem -Taylor’s and Laurent’s series – Contour integral in unit circle and semicircle (Excluding poles on real axis).		
TOTAL:45+15 = 60 Hours		
REFERENCES:		
1.G.B. Thomas and R. L.Finney, Calculus and Analytic geometry, 9 th Edition, Pearson, Reprint,2002.		
2. Erwin Kreyszig, Advanced Engineering Mathematics,9 th Edition, John Wiley & Sons, 2006.		
3.W.E.Boyce and R.C. DiPrima, Elementary Differential Equations and Boundary Value Problems, 9 th Edn., Wiley India, 2009.		
4. S.L.Ross, Differential Equations, 3 rd Ed.,Wiley India, 1984.		
5. J.W. Brown and R. V. Churchill, Complex Variables and Applications, 7 th Ed., Mc Graw Hill,2004.		
6. N. P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint,2008.		

2302EC201	ELECTRON DEVICES (Common to ECE and BME)										L	T	P	C
											3	0	0	3
PREREQUISITE:														
1. Semiconductor Physics														
COURSE OBJECTIVES:														
	1. To Describe the principal of operation, analysis and design of junction diode BJT and FET transistors and amplifier circuits													
	2. To understand various special devices and its applications													
	3. To explain the basics of PN junction fabrication													
COURSE OUTCOMES:														
At the end of this course, Students will be able to,														
CO1:	Describe about PN Junction diode and its characteristics													
CO2:	Examine the operation and characteristic features of BJT													
CO3:	Construct various JFET and MOSFET models													
CO4:	Test various Special semiconductor devices and its applications													
CO5:	Model various applications of PN diode													
CO6:	Construct different model of Regulators and SMPS													
Cos Vs Pos MAPPING:														
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11		
	CO1	3	3	2	3	1						2		
	CO2	3	3	2	3	1						2		
	CO3	3	3	2	3	2						2		
	CO4	3	2	3	2	2						2		
	CO5	3	2	3	2	1						2		
	CO6	3	2	3	2	1						2		
COs Vs PSOs MAPPING														
	COs	PSO1	PSO2	PSO3										
	CO1	2												
	CO2	2												
	CO3	2												
	CO4	2												
	CO5	2												
	CO6	2												
COURSE CONTENTS:														
Module I	SEMICONDUCTOR DEVICES											9 Hours		
PN junction diode, current equations, diffusion and drift current densities, forward and reverse bias characteristics, Transition and Diffusion capacitances, Switching characteristics, Breakdown in PN junction diodes.														
Module II	BIPOLAR JUNCTION TRANSISTOR											9 Hours		
NPN - PNP – Junctions - Early effect - Current equations – Input and Output characteristics of CE, CB, CC - Hybrid - π model - h-parameter model.														
Module III	FIELD EFFECT TRANSISTOR											9Hours		
JFETs – Drain and Transfer characteristics - Current equations- Current equation - Equivalent circuit model and its parameters. Characteristics of MOSFET,E-MOSFET,D-MOSFET, Current equation of MOSFET														
Module IV	SPECIAL SEMICONDUCTOR DEVICES											9Hours		
Zener diode - Varactor diode ,Tunnel diode, LASER diode, LDR,UJT, SCR, Diac, Triac, LED, LDR, Photo diode, Photo transistor.														
Module V	APPLICATIONS OF PN JUNCTION DIODE											9 Hours		
Rectifier-Half wave -Full-wave: Centre tapped and bridge rectifiers with resistive load -Analysis for Vdc and ripple voltage with C, C-L, L-C and C-L-C filters - Clippers and clampers - Zener diode and its application as regulator– Transistor voltage regulators: Series and shunt regulators -Switched mode power supply														
Total:45Hours														

REFERENCES:
1.Salivahanan. S and Sureshkumar.N, —Electronic Devices & Circuit, 3rd Edition, Tata McGraw- Hill, New Delhi, 2011, ISBN : 9781259006418.
2.Jacob Millman, Christos C. Halkias—Electronic Devices and Circuits 3rdEdition, McGraw Hill Education (India) Private Limited, 2010, ISBN :9780070700215
3.Allen Mottershead, —Electronic Devices and Circuits-An Introduction 1stEdition, PHI, New Delhi, 1990, ISBN: 9788120301245.
4.Electronic Devices and Circuits Theory, Boylsted, Prentice Hall Publications

2301GEX05	APPLIED DIGITAL LOGIC AND DESIGN					L	T	P	C		
	(Common to B.E - CSE, BME, B. Tech - IT, and AIDS)					3	0	0	3		
PREREQUISITE: Semiconductor physics											
COURSE OBJECTIVES:											
1. To present the fundamentals of digital circuits and simplification methods.											
2. To practice the design of various combinational and sequential digital circuits using logic gates.											
3. To introduce semiconductor memories and programmable logic devices.											
4. To practice the HDL programming for combinational and sequential circuits.											
COURSE OUTCOMES:											
At the end of this course, Students will be able to,											
CO1:		Use Boolean algebra, K-map and tabulation method to simplify Boolean functions.									
CO2:		Construct different combinational circuits using logic gates.									
CO3:		Develop different sequential circuits using logic gates and flip flops.									
CO4:		Compare different semiconductor memory devices.									
CO5:		Build programmable devices using logic gates.									
CO6:		Develop Verilog program for combinational and sequential circuits.									
COs Vs POs MAPPING:											
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	-	-	-	-	2	1	1
CO2	3	2	1	-	-	-	-	-	2	1	1
CO3	3	2	1	-	-	-	-	-	2	1	1
CO4	3	2	1	-	-	-	-	-	2	1	1
CO5	3	2	1	-	-	-	-	-	2	1	1
COs Vs PSOs MAPPING											
					COs	PSO1	PSO2	PSO2			
					CO1	1	1	2			
					CO2	1	1	1			
					CO3	1	2	1			
					CO4	1	1	1			
					CO5	1	1	1			
COURSE CONTENTS:											
Module I	BOOLEAN ALGEBRA AND LOGIC GATES								9 Hours		
Review of Number system – Boolean expression and minimization – Logic Gates and its implementation- Simplification of Boolean Functions using Boolean algebra, Karnaugh Map and Tabulation Method.											
Module II	COMBINATIONAL LOGIC								9 Hours		
Combinational Circuits – Analysis and Design Procedures – Circuits for Arithmetic Operations, Code Conversion – Decoders / Encoders – Multiplexers / Demultiplexers – Parity generators / checkers -Magnitude Comparator.											
Module III	SEQUENTIAL CIRCUITS								9 Hours		
Sequential logic-Basic latch-Flip - flops (SR, D, JK, T and Master-Slave) – Counters - Ripple counters - BCD and Binary - Synchronous counters, Registers - Shift registers - Registers, Hazards											
Module IV	MEMORY AND PROGRAMMABLE LOGIC								9 Hours		
Classification of memories (RAM, ROM, PROM, EPROM, EEPROM) - Programmable Logic Devices (PLA, PAL, FPGA) - Implementation of circuits using ROM, PLA, PAL.											
Module V	VERILOG HDL MODELING								9 Hours		
3 types of Verilog modeling (gate-level, dataflow, and behavioral)-Verilog programming for combinational and sequential circuits.											
TOTAL: 45 Hours											
REFERENCES:											
1. Morris Mano and Michael D. Ciletti, "Digital Design", 5th edition, Prentice Hall of India,2012											
2. Samir Palnitkar, "Verilog HDL", 2nd Edition, Pearson Education, 2003											
3. https://archive.nptel.ac.in/courses/108/105/108105132/ (Link for NPTEL/SWAYAM/MOOC Courses)											
4. https://www.vlab.co.in/broad-area-electronics-and-communications											

2301GEX02	ENGINEERING GRAPHICS	L	T	P	C
		2	1	0	3

PREREQUISITE:

1. Basic knowledge about geometry
2. Lettering and Dimensioning

COURSE OBJECTIVES:

- | |
|---|
| 1. To develop in students, graphic skills for communication of concepts, ideas and design of Engineering products |
| 2. To expose them to existing national standards related to technical drawings |

COURSE OUTCOMES:

At the end of this course, Students will be able to,

CO1:	Construct conic curves, involutes and cycloids
CO2:	Solve problems involving projection of points, lines and plane surfaces
CO3:	Draw the projection of a sectioned simple solids
CO4:	Draw the development of a sectioned simple solids
CO5:	Draw the orthographic, isometric projection of simple solids

COs Vs POs / PSOs MAPPING:

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	1	2	3				2	2	1			3
CO2	3	2	1	2	3				2	2	1			3
CO3	3	2	1	2	3				2	2	1			3
CO4	3	2	1	2	3				2	2	1			3
CO5	3	2	1	2	3				2	2	1			3

COURSE CONTENTS:

Module I	BASIC CONCEPTS OF TECHNICAL DRAWING AND PLANE CURVES	9 Hours
Importance of graphics in engineering applications – Use of drafting instruments – BIS conventions and specifications – Size, Scale, layout and folding of drawing sheets – Lettering and dimensioning. Basic Geometrical constructions, Curves used in engineering practices: Conics – Construction of ellipse, parabola and hyperbola by eccentricity method – Construction of cycloid – construction of involutes of square and circle – Drawing of tangents and normal to the above curves.		
Module II	PROJECTION OF POINTS, LINES AND PLANE SURFACES	9 Hours
Principal Planes-First angle projection-projection of points. Projection of straight lines (only First angle projections) inclined to both the principal planes - Determination of true lengths and true inclinations by rotating line method and traces. Projection of planes (polygonal and circular surfaces) inclined to both the principal planes by rotating object method.		
Module III	PROJECTION OF SOLIDS	9 Hours
Projection of simple solids like prisms, pyramids, cylinder and cone when the axis is inclined to one of the principal planes by rotating object method.		
Module IV	PROJECTION OF SECTIONED SOLIDS AND DEVELOPMENT OF SURFACES	9 Hours
Sectioning of above solids in simple vertical position when the cutting plane is inclined to the one of the principal planes and perpendicular to the other – obtaining true shape of section. Development of lateral surfaces of simple and sectioned solids – Prisms, pyramids cylinders and cones.		
Module V	ORTHOGRAPHIC AND ISOMETRIC PROJECTION	9 Hours
Visualization concepts–Representation of Three-Dimensional objects – Layout of views- Free hand sketching of multiple views from pictorial views of Objects. Isometric view - Prisms, pyramids, cylinders, cones. Principles of isometric projection – isometric scale –Isometric projections of simple solids and truncated solids - Prisms, pyramids, cylinders, cones- combination of two solid objects in simple vertical positions and miscellaneous problems.		
		TOTAL: 45 Hours
REFERENCES:		
1. Bhatt N.D. and Panchal V.M., Charotar Publishing House, 53rd Edition, 2019.		

2. Natrajan K.V., A Text Book of Engineering Graphics, Dhanalakshmi Publishers, Chennai, 2018.
3. Parthasarathy, N. S. and Vela Murali, “Engineering Drawing”, Oxford University Press, 2015.
4. Basant Agarwal and Agarwal C.M., “Engineering Drawing”, McGraw Hill, 2nd Edition, 2019.
5. Gopalakrishna K.R., “Engineering Drawing” (Vol. I&II combined), Subhas Publications, Bangalore, 27th Edition, 2017
6. Shah M.B., and Rana B.C., “Engineering Drawing”, Pearson Education India, 2 nd Edition, 2009.
7. Venugopal K. and Prabhu Raja V., “Engineering Graphics”, New Age International (P) Limited, 2008.

2301GEX03	PROBLEM SOLVING USING C	L	T	P	C
	(Theory cum Laboratory course)	2	0	4	4
PREREQUISITE:					
	1. Need some Mathematical Knowledge				
COURSE OBJECTIVES:					
1. To understand the constructs of C Language.					
2. To develop C programmes using arrays and strings					
3. To develop modular applications in C using functions and pointers					
4. To develop applications in C using structures and union					
COURSE OUTCOMES:					
On the successful completion of the course, students will be able to					
CO1:	Demonstrate the knowledge about the techniques used to solve problems in computing.				
CO2:	Build programmes using C constructs.				
CO3:	Design and implement applications using arrays and strings				
CO4:	Develop and implement modular applications in C using functions and pointers.				
CO5:	Develop programmes and applications in C using structures, union.				
COURSE CONTENTS:					
Module I	INTRODUCTION TO PROBLEM SOLVING TECHNIQUES			6 Hours	
Problem Solving Techniques – Algorithm – Flowchart – Pseudo code - Steps to convert Algorithm to Source code.Data Types – Constants – Keywords – Expressions – Type of Errors.					
Module II	BASICS OF C PROGRAMMING			6 Hours	
Structure of C programme – Pre-processor directives - Compilation process, Execution of source code. Operators and operator’s precedence – I/O statements – Sequence statements – Selection statements – Looping statements – Solve Numerical / Logical problems.					
Module III	ARRAYS AND STRINGS			6 Hours	
Introduction to Arrays: Declaration, Initialization – One dimensional array – Two dimensional array String operations: length, compare, concatenate, copy, upper case, lower case.					
Module IV	FUNCTIONS AND POINTERS			6 Hours	
Function prototypes - function definition, function call – Recursion: Binary search using recursive functions. Pointers – Pointer operators – Pointer arithmetic – Arrays and pointers – Parameter passing: Pass by value, Pass by reference.					
Module V	STRUCTURES, UNION AND FILE PROCESSING			6 Hours	
Structure - Nested structures – Pointer and Structures – Array of structures – Dynamic memory allocation – Union - Storage classes and Visibility, Types of file processing, Sequential access, Random access, Sequential access file, random access file, Comment line argument.					
TOTAL: 30 Hours					
LIST OF EXPERIMENTS:					

1. Prepare programmes in C to implement basic concepts in C language.	6 Hours
2. Produce C programmes to implement decision making and branching statements.	6 Hours
3. Use the concept of looping to implement C programmes.	6 Hours
4. Employ the concept of arrays to develop C programmes.	6 Hours
5. Experiment the concepts of strings using C.	6 Hours
6. Develop C programmes to perform code reusability using function.	6 Hours
7. Model programmes in C to implement pointers.	6 Hours
8. Build C programme to implement structures.	6 Hours
9. Build C programme to implement Union.	6 Hours
10. Mini Project: Using Structures, Functions & Pointers.	6 Hours

Hardware/software requirement

1. Desktop Systems 60 Nos
2. C Compiler

TOTAL: 60 Hours

TOTAL: 30+60=90 Hours

COs Vs POs / PSOs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1		1	1			2		1	1	3	1		
CO2	2	1		1	1			2		1	1	3	1		
CO3	3	2	3	1	1			2		1	1	3	1		
CO4	3	2	3	1	1			2		1	1	3	1		
CO5	3	2	3	1	1			2		1	1	3	1		

REFERENCES:

1. Yashwant Kanetkar, Let us C, 17th Edition, BPB Publications, 2020.
2. ReemaThareja, “Programming in C”, Oxford University Press, Second Edition, 2016.
3. Kernighan, B.W and Ritchie,D.M, “The C Programming language”, Second Edition, Pearson Education, 2015.
4. Byron S. Gottfried, “Schaum’s Outline of Theory and Problems of Programming with C”, McGraw-Hill Education, 1996.
5. https://onlinecourses.nptel.ac.in/noc23_cs53/course (Link for NPTEL/SWAYAM/MOOC Courses)
6. https://cse02-iiith.vlabs.ac.in/exp (Link for virtual Lab)
7. www.skillrack.com (Link for modern tool usage)

2304FLX02	JAPANESE LANGUAGE	L	T	P	C
		2	0	0	2
COURSE CONTENTS:					
Welcome aboard *ABK-AOTS DOSOKAI BENGALURU* *JLPT N5 integrated Course* covers 120 hours of intensive coaching, in preparation for JLPT exam + Revision for JLPT (certification by Japan Foundation , a world-wide standard) *Course content* 1 Japanese Scripts / alphabets : _Hiragana (native Japanese script) _Katakana (foreign words) _Kanji (Chinese derived script) 2. (bunpou) Grammar 25 lessons 3. (aisatsu) Greetings 4. (kaiwa) conversation through native Japanese enacted videos 5. (choukai) Listening to native Japanese conversion 6. (dokkai) Reading / comprehension & Revision - simulation (mock) tests ABK AOTS DOSOKAI					

2301TA201	தமிழரும் தொழில்நுட்பமும் / TAMILS AND TECHNOLOGY	L	T	P	C
		1	0	0	1

PREREQUISITE:

1. The Tamils living in different parts of the World need to keep in touch with the motherland and the mother tongue and be knowledgeable about their heritage in order to preserve their cultural identity and observe their traditional and cultural activities.

2. Recognizing this fact and for meeting the felt and emerging needs of the Tamil Communities and others interested in Tamil studies

COURSE OBJECTIVES:

1. Tamil Literature is way of a life. It focuses on the historical significance of ethics, moral culture in the Tamil context.

2. Tamil Modern literature emphasizes on the modern development of the behavioral, moral and ethical

3. Technology is the important key for a language and a new sector for the students to voice out for a social cause

COURSE OUTCOMES:

At the end of this course, Students will be able to,

CO1: Develop a spirit of patriotism.

CO2: Understand the plight of the people living in the society and Biological Struggles.

CO3: Remember the life style of the Sangam people and To recognize the heroic spirit of the ancient Tamil kings

CO4: Evaluate the quality and morals of local life through Tamil literature

CO5: Introducing the various Literary Genres and dramas and enable them to produce innovative ideas in modern literary theories

COs Vs POs & PSOs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-

Module I WEAVING AND CERAMIC TECHNOLOGY 3 Hours

Weaving Industry during Sangam Age–Ceramic technology–Black and Red Ware Potteries (BRW)

Graffition Potteries.

அலகு I நெசவு மற்றும் பானைத் தொழில்நுட்பம்: 3

சங்க காலத்தில் நெசவுத் தொழில் – பானைத் தொழில்நுட்பம் - கருப்பு சிவப்பு பாண்டங்கள் – பாண்டங்களில் கீறல் குறியீடுகள்.

Module II DESIGN AND CONSTRUCTION TECHNOLOGY 3 Hours

Designing and Structural construction House & Designs in house hold materials during Sangam Age Building materials and Hero stones of Sangam age -Details of Stage Constructions in Silappathikaram - Sculptures and Temples of Mamallapuram - Great Temples of Cholas and other worship places - Temples of Nayaka Period - Type study (Madurai Meenakshi Temple) - Thirumalai Nayakar Mahal – Chetti Nadu Houses, Indo-Saracenic architecture at Madras during British Period.

அலகு II வடிவமைப்பு மற்றும் கட்டிடத் தொழில்நுட்பம்: 3

சங்க காலத்தில் வடிவமைப்பு மற்றும் கட்டுமானங்கள் & சங்க காலத்தில் வீட்டுப் பொருட்களில் வடிவமைப்பு- சங்க காலத்தில் கட்டுமான பொருட்களும் நடுகல்லும் – சிலப்பதிகாரத்தில் மேடை அமைப்பு பற்றிய விவரங்கள் - மாமல்லபுரச் சிற்பங்களும், கோவில்களும் – சோழர் காலத்துப் பெருங்கோயில்கள் மற்றும் பிற வழிபாட்டுத் தலங்கள் – நாயக்கர் காலக் கோயில்கள் - மாதிரி கட்டமைப்புகள் பற்றி அறிதல், மதுரை மீனாட்சி அம்மன் ஆலயம் மற்றும் திருமலை நாயக்கர் மஹால் – செட்டிநாட்டு வீடுகள் – பிரிட்டிஷ் காலத்தில் சென்னையில் இந்தோ-சாரோசெனிக் கட்டிடக் கலை.

Module III	MANUFACTURING TECHNOLOGY	3 Hours
Art of Ship Building - Metallurgical studies - Iron industry - Iron smelting, steel - Copper and gold - Coins as source of history - Minting of Coins - Beads making - industries Stone beads - Glass beads - Terra-cotta beads - Shell beads/bone beads - Archeological evidences -Gems tone types described in Silappathikaram.		
அலகு III உற்பத்தித் தொழில் நுட்பம்:		3
கப்பல் கட்டும் கலை - உலோகவியல் - இரும்புத் தொழிற்சாலை - இரும்பை உருக்குதல், எஃகு - வரலாற்றுச் சான்றுகளாக செம்பு மற்றும் தங்க நாணயங்கள் - நாணயங்கள் அச்சடித்தல் - மணி உருவாக்கும் தொழிற்சாலைகள் - கல்மணிகள், கண்ணாடி மணிகள் - சுடுமண் மணிகள் - சங்கு மணிகள் - எலும்புத்துண்டுகள் - தொல்லியல் சான்றுகள் - சிலப்பதிகாரத்தில் மணிகளின் வகைகள்.		
Module IV	AGRICULTURE AND IRRIGATION TECHNOLOGY	3 Hours
Dam, Tank, ponds, Sluice, Significance of KumizhiThoompu of Chola Period, Animal Husbandry -Wells designed for cattle use - Agriculture and Agro Processing - Knowledge of Sea - Fisheries -Pearl - Conche diving - Ancient Knowledge of Ocean - Knowledge Specific Society		
அலகு IV வேளாண்மை மற்றும் நீர்ப்பாசனத் தொழில் நுட்பம்:		3
அணை, ஏரி, குளங்கள், மதகு - சோழர்காலக் குழுவித் தூம்பின் முக்கியத்துவம் - கால்நடை பராமரிப்பு - கால்நடைகளுக்காக வடிவமைக்கப்பட்ட கிணறுகள் - வேளாண்மை மற்றும் வேளாண்மைச் சார்ந்த செயல்பாடுகள் - கடல்சார் அறிவு - மீன்வளம் - முத்து மற்றும் முத்துக்குளித்தல் - பெருங்கடல் குறித்த பண்டைய அறிவு - அறிவுசார் சமூகம்.		
Module V	SCIENTIFIC TAMIL & TAMIL COMPUTING	3 Hours
Development of Scientific Tamil - Tamil computing - Digitalization of Tamil Books -Development of Tamil Software - Tamil Virtual Academy - Tamil Digital Library - Online Tamil Dictionaries -Sorkuvai Project.		
அலகு V அறிவியல் தமிழ் மற்றும் கணித்தமிழ்:		3
அறிவியல் தமிழின் வளர்ச்சி -கணித்தமிழ் வளர்ச்சி - தமிழ் நூல்களை மின்பதிப்பு செய்தல் - தமிழ் மென்பொருட்கள் உருவாக்கம் - தமிழ் இணையக் கல்விக்கழகம் - தமிழ் மின் நூலகம் - இணையத்தில் தமிழ் அகராதிகள் - சொற்குவைத் திட்டம்.		
TOTAL:15HOURS		
REFERENCES:		
1. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print) 2. Social Life of the Tamils - The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies. 3. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu) (Published by: International Institute of Tamil Studies). 4. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies.) 5. Keeladi - ‘Sangam City Civilization on the banks of river Vaigai’ (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu) 6. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Published by: The Author) 7. Porunai Civilization (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu) 8. Journey of Civilization Indus to Vaigai (R.Balakrishnan) (Published by: RMRL) – Reference Book.		

2301GEX51	COMPUTER PRACTICES LABORATORY										L	T	P	C
											0	0	2	1
PREREQUISITE:														
Basic knowledge about computer														
COURSE OBJECTIVES:														
1.	To be familiar with Computer Hardware Components and installation of software.													
2.	Make use of office package and to be familiar with the use of Office software.													
3.	To learn about searching, downloading, and storing contents in the Cloud Network.													
COURSE OUTCOMES:														
At the end of this course, Students will be able to,														
CO1	Perform assembling and disassembling of desktop machine with different peripheral and software installation and servicing.													
CO2	Simulate data using MS office for Presentation and Visualization.													
CO3	Use browsers for searching & accessing/storing the contents to/from cloud.													
LIST OF EXPERIMENTS:														
1. Familiarization of Computers & Computer Hardware Components														
2. Familiarization of major types of storage/memory technology														
3. Installing various operating systems including software download/installation, Familiarization of basic software/tools														
4. Working with MS-Office: MS Word, MS Excel, MS Power point														
5. Familiarization of Computer Shortcut keys														
6. Mini Project-1: Assemble your computer and install an Operating System														
7. Basics of Internet, Web browsers and Content Searching & accessing/storing the contents to/from cloud including Drop Box														
8. Familiarization of various types of security threats including virus														
9. Computer Ethics; Open Source way														
10. Mini Project-2: Document preparation using MS Word, Data Processing using MS Excel and Presentation using MS Power point														
TOTAL: 30 Hours														
COs Vs POs & PSOs MAPPING:														
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	2	2	-	-	-	2	2	-	-	1	1	1
CO2	3	3	2	2	2	-	-	-	-	1	-	1	1	1
CO3	3	3	2	1	-	-	-	2	-	-	-	1	1	1
HARDWARE/SOFTWARE REQUIREMENT														
1. Standalone Desktop Computers with Internet Connectivity														
2. Office Package														
3. Operating System Packages														
REFERENCES:														
1. Kevin Wilson, "Computer Hardware: The Illustrated Guide to Understanding Computer Hardware", 2021														
2. Kumar Bittu, "Mastering MS Office", 2020														
3. Ajay Mittal & Anitha Goel, "Computer Fundamentals and Programming in C", 2017														
4. https://nptel.ac.in/courses/106103068														
5. https://docs.oracle.com/cd/E19121-01/sf.x2100m2/819-6592-13/Chap1.html														
6. https://www.linkedin.com/learning/topics/microsoft-office														

2302EC251	ELECTRON DEVICES LABORATORY	L	T	P	C
	(Common to BME and ECE)	0	0	2	1

PREREQUEST:

1. Semiconductor Physics

COURSE OBJECTIVE:

1. To Describe the principal of operation, analysis and design of junction diode BJT and FET transistors and amplifier circuits.
2. To understand various special devices and its applications
3. To explain the basics of PN junction fabrication

COURSE OUTCOME:

- CO1** Describe about PN Junction diode and its characteristics
- CO2** Examine the operation and characteristic features of BJT
- CO3** Construct various JFET and MOSFET models
- CO4** Test various Special semiconductor devices and its applications

COs Vs POs & PSOs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	1	1	1				2	1	1	3		
CO2	3	2	1	1	1				2	1	1	3		
CO3	3	2	1	1	1				2	1	1	3		
CO4	3	2	1	1	3				2	1	1	2		

LIST OF EXPERIMENTS:

1. Construct and demonstrate Characteristics of PN Junction Diode, Zener diode	3 Hours
2. Construct and demonstrate Characteristics of Full Wave Rectifier using diode with and without filter	3 Hours
3. Construct and demonstrate input and output Characteristics of BJT in CE and CB Configuration	3 Hours
4. Construct and demonstrate Drain transfer Characteristics of JFET	3 Hours
5. Construct and demonstrate Characteristics of SCR	3 Hours
6. Construct and demonstrate Characteristics of UJT	3 Hours
7. Construct and demonstrate Characteristics of clippers and clampers	3 Hours
8. Simulation of any five experiments from the above in Multi-Sim software	3 Hours
9. Study and Construct series and shunt voltage regulator	3 Hours
10. Study and Construct SMPS using electronics devices	3 Hours
TOTAL:30Hours	

REFERENCES:

1. Salivahanan .S and Sureshkumar .N —Electronic Devices & Circuits, 3rdEdition, Tata McGraw- Hill, New Delhi, 2011, ISBN : 9781259006418.
2. Jacob Millman, Christos C. Halkias—Electronic Devices and Circuits, 3rdEdition, McGraw Hill Education (India) Private Limited, 2010, ISBN :9780070700215
3. Allen Mottershead —Electronic Devices and Circuits-An Introduction, 1stEdition, PHI, New Delhi, 1990, ISBN : 9788120301245
4. Electronic Devices and Circuits Theory, Boylsted, Prentice Hall Publications.
5. <https://www.youtube.com/watch?v=oqOG6XErAl8>
6. <https://www.youtube.com/watch?v=Kp-jS6NHsB8&list=PLF178600D851B098F>

2301GEX54	APPLIED DIGITAL LOGIC AND DESIGN LABORATORY	L	T	P	C
	Common to B.E-CSE, BME, B. Tech – IT and AIDS	0	0	2	1

PREREQUISITE: Basic mathematic skills

COURSE OBJECTIVES:

	1. To present the fundamentals of digital circuits and simplification methods.
	2. To practice the design of various combinational and sequential digital circuits using logic gates.
	3. To practice the HDL programming for combinational and sequential circuits.

COURSE OUTCOMES:

At the end of this course, Students will be able to,	
CO1:	Construct different combinational circuits using logic gates.
CO2:	Develop different sequential circuits using logic gates and flip flops.
CO3:	Build programmable devices using logic gates.
CO4:	Develop Verilog program for combinational and sequential circuits.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	-	-	-	-	2	1	-
CO2	3	2	1	-	-	-	-	-	2	1	-
CO3	3	2	1	-	-	-	-	-	2	1	-
CO4	3	2	1	-	3	-	-	-	2	1	-

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO2
CO1	1	1	2
CO2	1	2	1
CO3	2	2	1
CO4	2	2	2

LIST OF EXPERIMENTS

1. Verification of Boolean Theorems using basic gates
2. Design and implementation of half adder, half subtractor, full adder and full subtractor
3. Design and implementation of code converters
4. Design and implementation of multiplexer and de-multiplexer
5. Design and implementation parity generator/checker
6. Design and implementation counters
7. Design and implementation shift register
8. Develop and simulation of Verilog program for combinational circuits
9. Develop and simulation of Verilog program for sequential circuits

Hardware/software requirement

1. Digital trainer kit 10 Nos
2. Adequate numbers of IC's
3. Xilinx ISE (or) Altera Quartus II software

TOTAL: 30 Hours

REFERENCES:

1. Morris Mano and Michael D. Ciletti, "Digital Design", 5th edition, Prentice Hall of India, 2012
2. Samir Palnitkar, "Verilog HDL", 2nd Edition, Pearson Education, 2003
3. https://archive.nptel.ac.in/courses/108/105/108105132
4. https://www.vlab.co.in/broad-area-electronics-and-communications